

Arnav Garcha

PhD Candidate in Mechanical Engineering at Carnegie Mellon University

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EDUCATION

Carnegie Mellon University (CMU): Pittsburgh, PA

PhD, Mechanical Engineering – *in progress* (Advisor: Dr. Noelia Grande Gutiérrez) 2022–2027

MS, Mechanical Engineering, GPA: 4.0 2025

University of Waterloo: Waterloo, ON, Canada

BASc, Mechanical Engineering (GPA: 92%) – *Dean's Honors List with Distinction* 2017–2022

FELLOWSHIPS & SCHOLARSHIPS

Scholarship, Natural Sciences and Engineering Research Council of Canada – 120,000 CAD 2025–2027

Digital Health Fellowship, CMU School of Computer Science – \$90,000 2024–2025

Kothari International Experience Scholarship, University of Waterloo – 2,000 CAD 2020

Sanford Fleming Foundation Exchange Scholarship, University of Waterloo – 800 CAD 2020

President's Scholarship, University of Waterloo – 2,000 CAD 2017

HONORS & AWARDS

1st Place Poster in Computational Modeling and AI – University of Michigan, Engineering in Cardiovascular Medicine Workshop 2026

Institutional nominee for Schmidt Science Fellows Postdoctoral Fellowship – CMU 2026

Conference Travel Award – CMU Graduate Student Assembly 2026

Travel Award – 17th World Congress of Computational Mechanics, via USACM 2026

Finalist for Best Abstract – CMU Forum on Biomedical Engineering 2025

1st Place, Master's Student Competition (co-author) – ASME Summer Bioengineering Conference 2025

Sanford Fleming Foundation Academic Excellence Award (top of class) – University of Waterloo 2022

GRANTS

NSF ACCESS Project – High-performance simulations of arterial thrombosis (lead PI, \$4,000) 2026

PUBLICATIONS

Published Journal Articles

1. Jolly, T., **Garcha, A.**, Grande Gutiérrez, N. Impact of Microvascular Resistance on Coronary Hemodynamic Diagnostic Indices: A Multiscale Simulation Study. *Annals of Biomedical Engineering*. (2026). doi:10.1007/s10439-026-04378-1
2. **Garcha, A.**, Grande Gutiérrez, N. Scale-dependent mechanisms of near-wall platelet enrichment in arterial flows. *Physics of Fluids* 38, 041904 (2026). *Editor's Pick*. doi:10.1063/5.0324694
3. **Garcha, A.**, Grande Gutiérrez, N. Sensitivity of coronary hemodynamics to vascular structure variations in health and disease. *Scientific Reports* 15, 3325 (2025). doi:10.1038/s41598-025-85781-x

Conference Proceedings (Refereed)

1. **Garcha, A.***, Cheney, N., Kumar, R., Hunte, I., Allen, A. (2026). Sharing our voice: The value of incorporating student leaders in institutional community engagement during shifting climates. *American Society for Engineering Education Annual Conference*. June 2026.

Articles in Preparation

- **Garcha, A.**, Grande Gutiérrez, N. Arterial-scale thrombosis model of platelet adhesion, activation, and aggregation.

- **Garcha, A.**, Abdelmohsen, L., Najmi, A., Xu, W., Grande Gutiérrez, N. Best Practices for Physiological, Patient-Specific Boundary Condition Selection for Vascular Computational Fluid Dynamics Simulations.
- Najmi, A., **Garcha, A.**, Grande Gutiérrez, N. Transport patterns for nutrient delivery in a computational multiscale placentone model.
- Xu, W., **Garcha, A.**, Robles, J., Karthikeyakannan, M., Jolly, T., Grande Gutiérrez, N. Accelerating cardiovascular CFD simulations using a GNN super-resolution framework.

CONFERENCE PRESENTATIONS (* = PRESENTER)

1. **Garcha, A.***, Grande Gutiérrez, N. Toward a clinical thrombosis model: Platelet adhesion, activation, and aggregation in arteries. *17th World Congress on Computational Mechanics*, Munich, Germany, July 2026. (Oral)
2. **Garcha, A.**, Grande Gutiérrez, N*. Toward an arterial thrombosis model: Simulating platelet adhesion, activation, and aggregation. *9th International Conference on Computational and Mathematical Biomedical Engineering*, Kobe, Japan, June 2026. (Oral)
3. **Garcha, A.***, Grande Gutiérrez, N. Simulating platelet transport in arterial flows. *Carnegie Mellon Forum on Biomedical Engineering*, September 2025. (Virtual poster/presentation)
4. **Garcha, A.***, Grande Gutiérrez, N. Investigating platelet adhesion in arterial-scale flows using a hybrid continuum-discrete model. *20th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering*, Barcelona, Spain, September 2025. (Oral)
5. **Garcha, A.***, Grande Gutiérrez, N. Discrete platelet wall-attraction model for arterial-scale flows. *ASME Summer Bioengineering Conference*, Santa Ana Pueblo, NM, June 2025. (Poster)
6. Jolly, T.*, **Garcha, A.**, Grande Gutiérrez, N. Microvascular resistance influence on diagnostic indices of coronary hemodynamics. *ASME Summer Bioengineering Conference*, Santa Ana Pueblo, NM, June 2025. (Poster)
7. **Garcha, A.***, Grande Gutiérrez, N. Revisiting the relevance of platelet margination in arteries: A computational investigation. *APS Division of Fluid Dynamics*, Salt Lake, UT, November 2024. (Oral)
8. **Garcha, A.***, Grande Gutiérrez, N. Plaque length and stenosis influence instantaneous wave-free ratio and wall shear stress. *ASME Summer Bioengineering Conference*, Lake Geneva, WI, June 2024. (Oral)
9. **Garcha, A.***, Grande Gutiérrez, N. Evaluating the sensitivity of coronary artery hemodynamics to anatomical changes in angles, branch positions, and diameter ratios. *APS Division of Fluid Dynamics*, Washington, DC, November 2023. (Oral)
10. **Garcha, A.***, Grande Gutiérrez, N. Investigating the impact of coronary vascular structure diversity on local and global hemodynamics. *BMES Annual Meeting*, Seattle, WA, October 2023. (Oral)

OTHER PRESENTATIONS

1. **Garcha, A.***, Grande Gutiérrez, N. A complete framework for simulating thrombosis initiation in arterial vasculatures. *Summer BME Workshops at the University of Michigan; Engineering in Cardiovascular Medicine*, Ann Arbor, MI, August 2026. (Poster)
2. **Garcha, A.***, Grande Gutiérrez, N. Arterial-scale thrombosis model. *CMU PhD Poster Session*, Pittsburgh, PA, February 2026. (Poster)
3. **Garcha, A.***, Grande Gutiérrez, N. Hybrid discrete-continuum platelet adhesion model. *CMU PhD Poster Session*, Pittsburgh, PA, March 2025. (Poster)
4. **Garcha, A.***, Grande Gutiérrez, N. Considering vascular structure diversity for risk stratification of coronary artery disease. *CMU PhD Poster Session*, Pittsburgh, PA, March 2024. (Poster)

RESEARCH EXPERIENCE

Biomedical Flow Simulation and Multiscale Modeling Lab: CMU <i>Graduate Research Assistant</i>	2022–Present
XL Aero – Aeronautics Engineering Consultancy: British Columbia, Canada <i>R&D Engineering Trainee</i>	2020–2021
Materials in Biological Systems Lab: University of Waterloo <i>Undergraduate Research Assistant</i>	2019–2022

TEACHING & MENTORING

CMU – Teaching Assistant, undergraduate fluid mechanics (CMU 24-231)	2025, 2026
CMU – Master’s Student Mentorship (Tej Jolly, now PhD student at Oxford University)	2024–2026
CMU – Visiting Student Mentorship (Julen Martin Robles, now PhD student at CMU)	2025
CMU – Summer Undergraduate Research Assistantship (SURA) Project Mentor	2023, 2026

SERVICE

Physics of Fluids – Reviewer	2026
ASEE Annual Conference – Reviewer, student paper competition	2026
ASME Summer Biomechanics Conference – Judge, bachelor’s student poster competition	2025
CMU College of Engineering – Student Member, Dean Search Committee	2025
CMU Counseling and Psychological Services – Student Advisory Board Member	2024–2026
CMU College of Engineering – Community Engagement Student Council Member	2023–2026

COMMUNITY OUTREACH

CMU National Biomechanics Day – Co-director of outreach events in the local region	2025–Present
CMU High School Summer Engineering Camp – Instructor for biofluid mechanics	2024
CMU Gelfand Center for Education Outreach – Co-instructor for biofluid mechanics	2023

TECHNICAL SKILLS

Scientific Programming: Python, C++, C, Bash, FORTRAN, MATLAB, L^AT_EX, PyTorch
Compute Frameworks: High-performance computing (HPC), MPI, OpenMP, UNIX, Git, Docker
Computer-Aided Design: SolidWorks, AutoCAD, Onshape, VTK
Numerical Simulation Software: ANSYS, COMSOL, SolidWorks